Course outline

About NPTEL

How does an NPTEL online course work?

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- Data Privacy
- Differential Privacy
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Week 6: Assignment 6

The due date for submitting this assignment has passed.

Due on 2024-09-(

As per our records you have not submitted this assignment.

1) Which of the following methods is the best method to efficiently protect the data to preserve the privacy of the users?

- Anonymization
- Cryptographical Solution
- Statistical solution
- Data Compression
- Data Duplication

No, the answer is incorrect.
Score: 0
Accepted Answers:
Statistical solution

2) Between a randomized response (with epsilon>0) and a fair coin toss response, which algorithm would you use to preserve privacy but better utility?

- Randomized response because the chance of falsehood is 50%
- Randomized response because the chance of truth is greater than 50%
- Randomized response because the chance of falsehood is greater than 50%
- Coin toss response because the chance of falsehood is 50%
- Coin toss response because the chance of truth is greater than 50%
- Coin toss response because the chance of falsehood is greater than 50%

No, the answer is incorrect.
Score: 0
Accepted Answers:
Randomized response because the chance of truth is greater than 50%

3) Consider the equation in the context of privacy guarantees (The notations used are the same as used during the lecture).

$$P(RR(x') = b)^* e^{-\varepsilon} \leq P(RR(x) = b) \leq P(RR(x') = b)^* e^{\varepsilon}$$

To maximize the privacy gains, which of the following values should be changed and how?

- ε should be maximum for privacy, ε should be minimum for utility
- ε should be minimum for privacy, ε should be minimum for utility
- ε should be maximum for privacy, ε should be maximum for utility
- ε should be minimum for privacy, ε should be maximum for utility
- ε is unrelated

No, the answer is incorrect.
Score: 0
Accepted Answers:
ε should be minimum for privacy, ε should be maximum for utility

4) Consider the below values:
$X = x1, x2, \ldots . xN$ is the truth of an experiment
$Y = y1, y2, \ldots . yN$ is the revealed values instead of the truth
To identify the average of truth, Y as an estimator cannot be used for the process by which it was obtained. You derive new values Z where $Z = z1, z2, \ldots . zN$ from Y which are better estimators of X. How do you arrive at the values Z?

- Removing the bias from Y introduced through the random process
- Adding the bias to Y removed through the random process
- Removing the variance from Y introduced through the random process
- Adding the variance to Y removed through the random process

No, the answer is incorrect.
Score: 0
Accepted Answers:
Removing the bias from Y introduced through the random process

5) If ε is fixed, given a privacy guarantee, to improve the utility, which of the following values can be modified?

- Increase the number of experiments
- Increase the amount of randomness

- Increase the amount of bias introduced in the random process
- Increase the amount of variance introduced in the random process

No, the answer is incorrect.
Score: 0

Accepted Answers:
Increase the number of experiments

6) Identify the equation for the ε-differential mechanism (The notations used are the same as used during the lecture):

- $\frac{P(M(x)\in S)}{P(M(x')\in S)} \leq e^{\varepsilon}$
- $\frac{P(M(x')\in S)}{P(M(x')\in S)} \leq e^{\varepsilon}$
- $\frac{P(M(x)\in S)}{P(M(x)\in S)} \leq e^{\varepsilon}$
- $\frac{P(M(x)\in S)}{P(M(x')\in S)} \geq e^{\varepsilon}$
- $\frac{P(M(x')\in S)}{P(M(x')\in S)} \geq e^{\varepsilon}$
- $\frac{P(M(x)\in S)}{P(M(x)\in S)} \geq e^{\varepsilon}$

No, the answer is incorrect.
Score: 0

Accepted Answers:
$\frac{P(M(x)\in S)}{P(M(x')\in S)} \leq e^{\varepsilon}$

7) Identify the correct scenario in the case of differential privacy

- Trust the curator; Trust the world
- Do not trust the curator; Trust the world
- Trust the curator; Do not trust the world
- Do not trust the curator; Do not trust the world

No, the answer is incorrect.
Score: 0

Accepted Answers:
Trust the curator; Do not trust the world

8) Identify all the values representing sensitivity in a laplacian mechanism where the function under consideration is an average of n binary {0,1} (The notations used are the same as used during the lecture).

- $\frac{1}{n}$
- $\frac{1}{n}\ |x_n' - x_n|$
- ε
- $\frac{\varepsilon}{n}$
- $\frac{-1}{n}$
- $\frac{\Delta}{\varepsilon}$

No, the answer is incorrect.
Score: 0

Accepted Answers:
$\frac{1}{n}$
$\frac{1}{n}\ |x_n' - x_n|$

9) Identify the distribution from which the noise is derived in a laplacian mechanism. The representation is of the form Laplacian(a,b) where mean and b is the spread parameter. (The notations used are the same as used during the lecture)

- laplacian $(1, \frac{\Delta}{\varepsilon})$
- laplacian $(\frac{\Delta}{\varepsilon}, 0)$
- laplacian $(\frac{\Delta}{\varepsilon}, 1)$
- laplacian $(0, \frac{\Delta}{\varepsilon})$
- laplacian $(1, 1)$
- laplacian $(0, 0)$

No, the answer is incorrect.
Score: 0

Accepted Answers:
laplacian $(0, \frac{\Delta}{\varepsilon})$

10) Higher privacy guarantees can be achieved in which of the following scenarios? Identify all the possible scenarios.

- Epsilon should be high
- Inverse Sensitivity should be high

- [ ] Variance should be high
- [ ] Noise should be high
- [ ] Utility should be high

No, the answer is incorrect.
Score: 0

Accepted Answers:
Variance should be high
Noise should be high

11) Identify the deviation of the value from the truth in the scenario of a laplacian mechanism. (The notations used are the same as used du lecture).

- $o(\frac{1}{\varepsilon n})$
- $o(\frac{n'}{\varepsilon n})$
- $o(\frac{\varepsilon}{n})$
- $o(\frac{e}{\varepsilon n})$
- $o(\varepsilon n)$

No, the answer is incorrect.
Score: 0

Accepted Answers:
$o(\frac{1}{\varepsilon n})$

12) In the scenario of a privacy-utility trade-off, for fixed privacy, the number of samples required for a particular utility varies between the L mechanism and Randomized response is different by what factor?

- Constant factor
- Linear factor
- Exponential factor
- Logarithmic factor
- Quadratic factor

No, the answer is incorrect.
Score: 0

Accepted Answers:
Quadratic factor